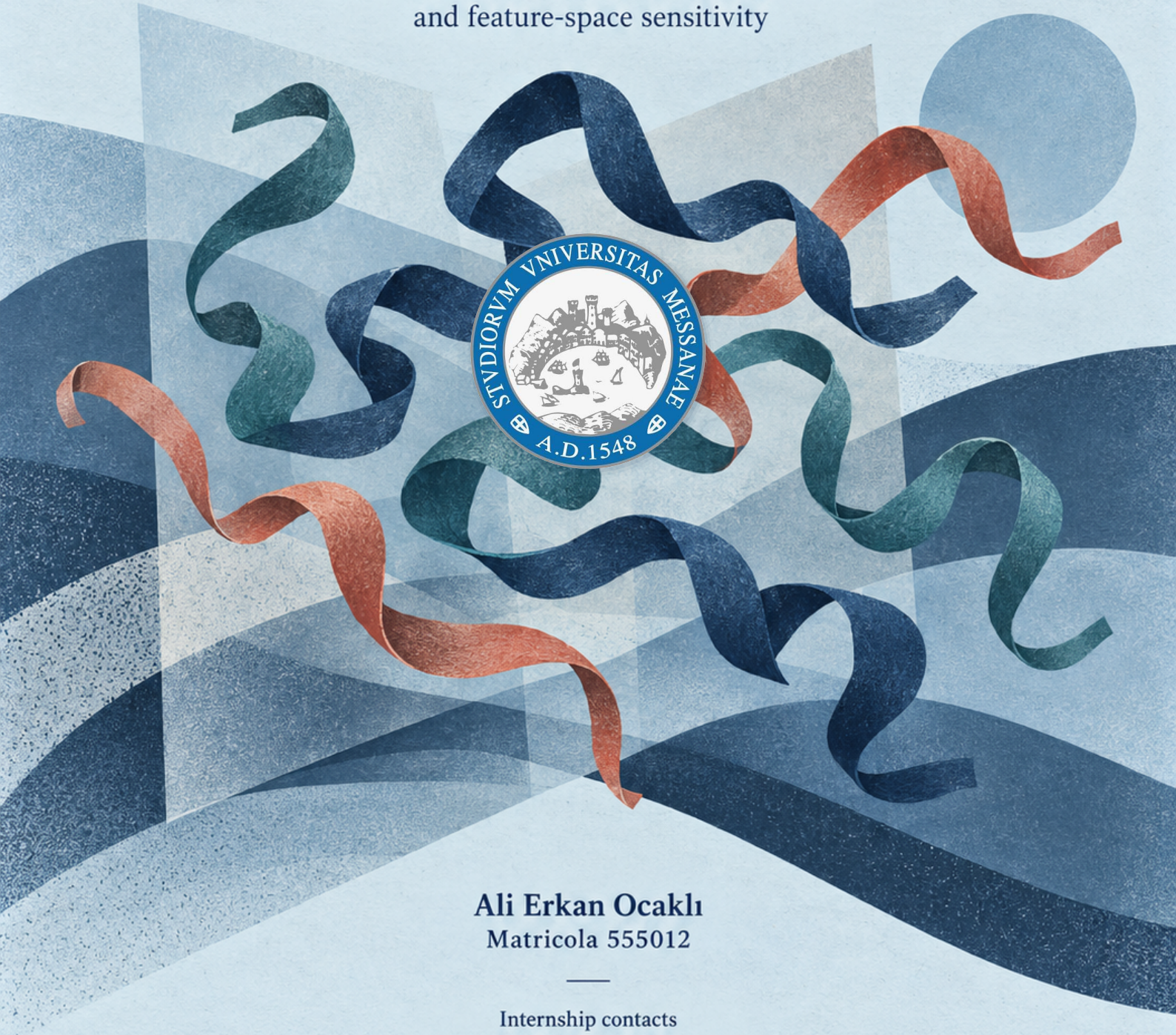
UNIVERSITY OF MESSINA
Data Analysis L-31

CURRICULAR INTERNSHIP REPORT

RobustCluster-Aβ

Reproducible analysis of amyloid-β40 conformational descriptors

Scientific code audit, sampling diagnostics and feature-space sensitivity



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Abstract

This internship project examines how the choice of structural measurements affects clustering of amyloid-β40 configurations. The starting material was a descriptor table and an analysis script associated with the workflow of Cadenelli and colleagues. The work had two parts: checking and reorganising that analysis, and testing whether solvent-accessible surface area (SASA) could be left out when radius of gyration was already included.

The table contains 300,003 saved observations from three replicas over the source-labelled 1–3 microsecond interval. The code audit found errors in parsing and several steps used to form and describe clusters. The revised Python package includes numerical tests and records the settings of each run. A complete free-energy grid was also checked. Its energy units and method of construction could not be verified, so the analyses reported here do not estimate equilibrium populations.

Rg and SASA have a pooled correlation of 0.848, and the first two standardized principal components explain 88.1% of feature variance. Gaussian mixture models were then fitted with and without SASA, using the same sample with equal numbers of frames from each replica. Five component counts and five starting seeds gave fifty fits, all of which met the numerical convergence criterion. With three components, the median Adjusted Rand index between the two sets of assignments was 0.864; with six, it was 0.457. Removing SASA therefore had a different effect at different component counts, despite its strong correlation with Rg.

All replica–descriptor series failed at least one reliability check for the correlation-time estimate. The mixture comparison also used only one fitting sample, and its evaluation included the fitting rows. These limits mean that the results describe grouping in the supplied biased sample, not physical metastable states or unbiased kinetic rates.

The code, saved results and report document what was checked and what remains unfinished. Reproducing equilibrium conformational populations still requires verified information about the free-energy surface and a separate calculation.

Keywords: amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; density peaks; statistical reweighting; reproducibility.

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Abbreviations and evidence categories

Term	Meaning
ACF	Autocorrelation function
ARI	Adjusted Rand index
CV	Collective variable
GMM	Gaussian mixture model
DPC / DPA	Density Peak Clustering / Density Peaks Advanced
ESS	Effective sample size; estimator and assumptions stated explicitly
FES	Free-energy surface
IDP	Intrinsically disordered protein or peptide
MD	Molecular dynamics
PCA	Principal component analysis
SASA	Solvent-accessible surface area
SEM	Standard error of the mean
WT-MetaD	Well-tempered metadynamics

The report separates the published method, its implementation, corrections to the supplied code, additional robustness experiments, and future work. A software test checks how the code behaves on a specified example; it does not establish a biological result. An analysis marked BLOCKED lacks required inputs or information about their origin. Learning topics in the internship programme are also distinguished from work that has a recorded output.

1 Introduction and internship objectives

1.1 Biological and computational motivation

Clustering groups peptide configurations according to a small set of measured properties. The choice of properties matters for amyloid-β monomers: configurations of similar size can differ in secondary structure, solvent exposure and contacts. Intrinsically disordered peptides still have structural organization, even when no single conformation describes the ensemble well [1]. The task is to choose a useful summary without losing sight of the molecular details it leaves out.

Cadenelli and colleagues compared monomeric Aβ40 and Aβ42 ensembles using the same simulation and analysis protocol [2]. Only Aβ40 descriptors were supplied for this internship, so this report examines how those descriptors are analysed. It does not make a new comparison between the two peptides. Energy-landscape studies help explain the interest in monomer ensembles, but grouping configurations is not itself a measurement of aggregation [3].

1.2 Internship context and objectives

This work forms part of the Data Analysis L-31 internship at the University of Messina, with Andrea Ciccolo and Valeria Conti Nibali as internship contacts. The programme covered protein structure and intrinsic disorder, followed by force fields, molecular dynamics and clustering. The final assignment asked for an independent method or analysis using knowledge from the degree programme.

The work began by checking the supplied script and data against the published method, then implementing the required corrections. The descriptor analysis raised a second question. Radius of gyration and SASA are correlated, but would the clusters stay similar if SASA were removed? The independent experiment tests this by fitting Gaussian mixtures with different numbers of components and different initializations. This comparison can use the available descriptors without equilibrium weights.

1.3 Contributions and evidence boundaries

The completed work includes the code audit, tested analysis routines, descriptor and sampling checks, validation of the FES grid, and the Gaussian-mixture comparison with and without SASA. This last experiment is the original analysis developed for the report. Reproduction of the published clustering method remains a separate, unfinished task because the FES energy units and the way it was constructed are still unknown.

Each calculation saves its inputs, settings, software versions and outputs. A reader can use these records to trace a plot back to its source table and check whether the calculation completed. This follows guidance on reusable workflows and records that can be checked by software [4–6]. The practical aim is simple: someone reading the report should be able to see how each result was obtained.

2 Scientific and theoretical background

2.1 Amyloid-β40 and intrinsically disordered ensembles

An ensemble describes the conformations a molecule can adopt and the statistical weight of each one. For an intrinsically disordered protein (IDP), a representative structure shows only one member of the ensemble, whereas an ensemble-averaged measurement reflects many conformations. Global descriptors summarize this information but lose some structural detail. Two configurations can have the same radius of gyration and solvent-accessible area but different residue contacts. Clustering these descriptors groups conformations by the measured similarities; it does not identify a unique atomic fold.

Experimental data can help constrain an ensemble, although NMR and SAXS measurements generally average over conformations. Borthakur and colleagues describe a maximum-entropy procedure that combines simulations with experimental data to constrain IDP ensembles [7]. Their study offers a possible independent comparison for this project, provided that the structural data, measured quantities and statistical weights are compatible.

2.2 Molecular dynamics and conformational ensembles

Molecular dynamics generates configurations by numerically integrating the equations of motion under a specified potential-energy function. Force-field parameters, the solvent model, thermodynamic conditions and sampling all affect the resulting ensemble. The primary study used GROMACS, CHARMM36m and PLUMED [2, 8–10]. This project starts from the descriptor table produced after those simulations.

In an equilibrium canonical ensemble, an observable $A(q)$ has expectation proportional to the integral of $A(q)\exp[-\beta_T U(q)]$, where q denotes coordinates and U is potential energy. For energies per molecule, $\beta_T = 1/(k_B T)$; for the molar energies used in the implementation, the consistent factor is $1/(RT)$. Mixing these energy conventions introduces a scale error even when the reweighting code is numerically stable.

2.3 Sampling limitations

Saving configurations more frequently adds rows to a trajectory without necessarily adding independent information. Nearby frames may remain in the same slowly changing region of descriptor space. Under suitable assumptions, statistical inefficiency measures how this dependence increases the variance of an estimate. It cannot account for a region of the ensemble that was never sampled. Even a stable overall mean can conceal changes in the relative populations of different structural groups.

Assessing equilibration and sampling quality requires more than a raw frame count [11, 12]. Here, the trajectory was also generated under a changing bias. Correlation statistics describe dependence in that recorded process; they do not give unbiased relaxation times, transition rates, or kinetic states. The reproduction retains the published analysis interval. The extension examines the statistics within that interval without changing its boundaries.

2.4 Well-tempered metadynamics

WT-MetaD adds a history-dependent potential along selected CVs. Its tempering parameter is conventionally expressed as $\gamma = (T + \Delta T)/T > 1$. In the long-time regime, the bias approaches a scaled negative free energy, up to an additive constant [13]. This relationship motivates the final-bias approximation used in the primary paper.

The bias encourages exploration but also changes how often conformations are sampled and the forces acting on the peptide. Population estimates therefore need a statistical correction, and the recorded time cannot be read directly as an unbiased physical clock. Thermodynamic reweighting corrects the distribution under stated assumptions; recovering kinetic information is a separate problem.

2.5 Gaussian hills and bias construction

For CV vector s, hills deposited at times t_k generate a bias of the form

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{2.1}$$

In WT-MetaD, hill heights decrease with the accumulated bias as $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B \Delta T)]$, using molecular energy units. For molar units, k_B is replaced by R. The convention must agree with the simulation software and energy units. Equation 2.1 explains how the bias is constructed. It has not been reconstructed here because no project HILLS file is available.

PLUMED distinguishes the instantaneous bias output, the time-dependent offset, and their difference. Its `metad.rbias` component equals `metad.bias` minus `metad.rct`; the latter represents $c(t)$ [14]. A column named `metad.rbias` is therefore potentially useful for a separate time-dependent reweighting analysis. Its name alone does not establish that the offset was computed with the required settings or that its units are known.

2.6 Free-energy surfaces

For a CV distribution $p(s)$, free energy is $F(s) = -\beta_T^{-1} \log p(s) + C$, where C sets an arbitrary reference. This surface describes the chosen CVs. Integrating over the remaining coordinates can hide structural differences, so a two-dimensional minimum does not uniquely specify a three-dimensional conformation.

PLUMED's `sum_hills` supports reconstruction from accumulated hill records and distinguishes free-energy output from negative-bias output [15]. That distinction matters because the WT scaling factor must not be applied twice. The recovered complete file labels its energy field `file.free`, but the energy unit, generating commands, and replica-combination procedure are still needed for a verified final-bias calculation.

2.7 Statistical reweighting

For a quasi-static bias, the correction from a biased distribution to the corresponding unbiased distribution has weights proportional to $\exp(\beta_T V)$. Combining this with the WT relationship yields

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2.2}$$

The free-energy zero does not affect the normalized result: adding a constant to F adds the same offset to all relevant log weights, which cancels on normalization. Equation 2.2 implements the primary paper's final-bias approximation. It assumes that the analyzed interval can be treated as quasi-stationary, with a sufficiently stable bias for this correction to apply.

Alternative MetaD reweighting approaches account explicitly for changing bias normalization [16, 17]. For verified PLUMED offset-corrected output, the optional comparison uses $\ell_i = \mathrm{rbias}_i/(RT)$ [18]. These two estimators are not interchangeable labels for the same computation. The current repository keeps their configurations and resulting outputs separate.

2.8 Structural descriptors

The baseline uses n_α, n_β, R_g, and total SASA. In the primary study, the first two are scaled continuous secondary-structure CVs, intended to approximate residue counts. They should not be interpreted as exact integer DSSP assignments. DSSP itself classifies local structural patterns using geometrical and hydrogen-bond information [19]; reproducing such assignments requires coordinates, which are absent here.

Radius of gyration summarizes global compaction; SASA measures solvent exposure. They may be correlated while still distinguishing different conformations within a cluster. Apolar and polar SASA,

intrapeptide hydrogen bonds, and peptide–solvent hydrogen bonds provide further descriptors after baseline clustering. Their exact definitions must be recovered from the original analysis settings, including whether the two SASA components add up to total SASA.

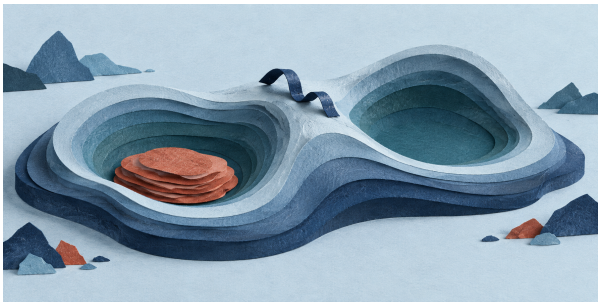
2.9 Density-based conformational clustering

Density-peak methods seek centers that have both high local density and a large distance from denser points. The primary study applies this idea repeatedly to find acceptable clusters across random subsets and density scales [2]. A silhouette coefficient compares how close a point is to its own cluster with how close it is to other clusters [20]. It measures geometric separation and does not establish thermodynamic metastability.

Consensus then groups local clusters by similarities in their descriptor distributions. DPA offers a distinct, statistically motivated density-topography approach [21]. A DPA benchmark would test whether an alternative treatment of density produces similar families. Likewise, weighted ShapeGMM addresses structural ensembles in a coordinate-based representation [22]; its applicability here depends on obtaining valid structural frames.



(a) Conformational diversity



(b) An evolving sampling landscape

Figure 1: Conceptual editorial illustrations. Open paper ribbons evoke different conformations of a flexible peptide; layered deposits evoke the accumulation of metadynamics bias. Neither image depicts molecular coordinates, secondary-structure assignments, a measured FES or a physical trajectory. The qualitative ideas are formalized in the surrounding text and eq. (2.1). Generated-art provenance is documented in Appendix D.

3 Data and computational methodology

3.1 Data provenance

The input is a 10-column text table with 300 003 numerical rows. Its SHA256 and the hash of the original script are recorded in the inventory. The original materials are retained unchanged. The table contains scalar descriptors and a column labelled `metad.rbias`. Contact maps or molecular renderings would additionally require atomic coordinates and a verified mapping from each row to its structural frame.

The paper describes the method to reproduce, while the supplied script shows its implementation. Neither alone establishes that an analysis completed; that requires execution records. Scientific V2 and its changelog were unavailable, and the method map records this gap.

3.2 Replica segmentation

Replica boundaries are detected where the first column decreases. Parsing requires exactly three segments, strictly increasing times within each segment, finite numerical entries, and a recognized header. Extra segments, duplicate times, malformed rows, or missing interval coverage fail validation rather than being silently skipped.

Before selecting or shuffling frames, the parser saves their identities: zero-based source row, one-based file line, replica ID and zero-based frame index within the original segment. These identifiers let clustering comparisons match the same observations across runs. Comparing arrays by their storage order alone could mistake reordered frames for changed assignments.

3.3 Analysis interval

The baseline interval includes both endpoints from 1 000 000 to 3 000 000, with time interpreted in picoseconds as labelled in the source. Each replica contributes 100 001 frames separated by 20 ps. The original script and published protocol support this interpretation, but the original simulation and analysis metadata are absent. The conversion to the supplied 1–3 µs window is therefore recorded as an assumption.

The published workflow excludes the first microsecond. The supplied table already starts at that boundary. It allows the analysis interval to be checked, but not the preceding equilibration period or convergence over the full trajectory.

3.4 Descriptor definitions

The clustering columns are n_α, n_β, R_g in nm, and SASA in nm^2, with units taken from the original script. The parser explicitly maps differences in header spelling, including the original `HBONNDSPS` spelling. It preserves `metad.rbias` as a separate field rather than treating it as an interpolated free energy.

The FES and the clustering algorithm use different coordinates. FES queries apply the paper's descriptor conversions, $\alpha = n_\alpha/1.1$ and $\beta = n_\beta/2.5$, whereas clustering uses normalized features. Passing min–max-scaled values to the FES would select different physical CV locations. Separate modules and saved parameters keep these transformations distinct.

3.5 Feature normalization

For each baseline descriptor x_f, min–max scaling uses all replicas within the analysis window: $z_f = (x_f - a_f)/(b_f - a_f)$, where a_f and b_f are the pooled minimum and maximum. Both values are saved. A constant feature would make the denominator zero, so validation rejects it rather than silently removing the descriptor.

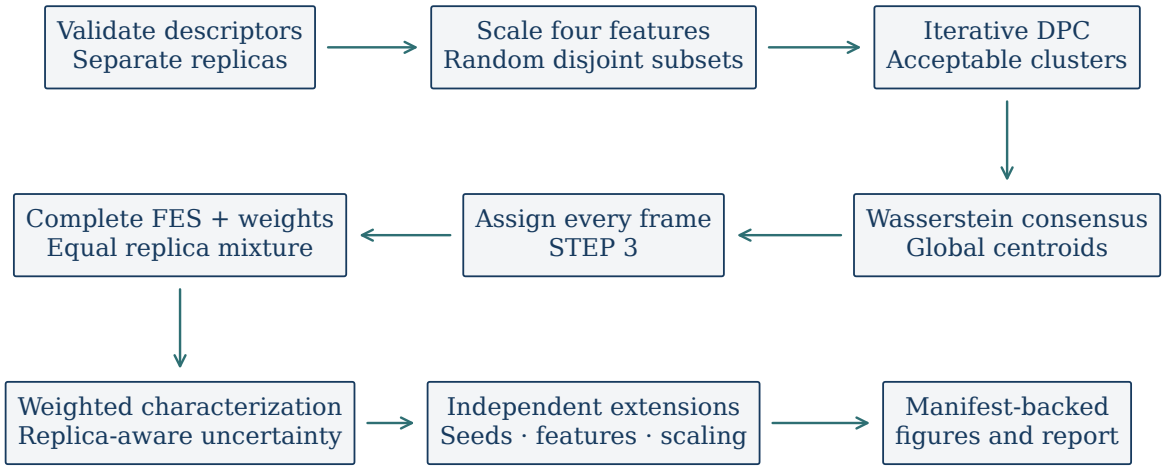


Figure 2: Implemented analysis design. The primary branch separates geometric cluster discovery from statistical reweighting. Robustness experiments rerun designated stages independently. The FES grid has been validated, while empirical weighting remains blocked by energy-unit and construction provenance; this workflow diagram documents implementation, not successful completion of every box.

The pooled transform gives every replica and discovery group the same coordinate system. If each group were scaled independently, an observation's numerical position would depend on the group to which it had been randomly assigned. The baseline therefore retains the source's common min–max scaling. Standardization belongs to the separate GMM protocol; the planned DPC scaling comparisons remain unperformed.

3.6 STEP 1: iterative Density Peak Clustering

The workflow in fig. 2 begins by dividing frames into random, nonoverlapping subsets. Production group size is approximately 30 000. If the final subset contains at most ten frames, those frames are redistributed over the preceding groups without losing their identities. Local random-number generator (RNG) streams are derived from a recorded master seed. This removes the limit imposed by the original fixed seed list, but produces different random draws from the original global-RNG implementation [23].

Local density is the number of neighbors within d_c, including the point itself as in the supplied code. To handle equal densities reproducibly, points are ordered by density and then by local frame position. The separation δ_i is measured to the closest earlier point in that order; the first point uses the maximum pairwise distance. Ranking $\rho_i\delta_i$ then identifies candidate centers that combine density with separation. The tie rule matters because an earlier point need not be strictly denser.

For each cutoff, the algorithm tries candidate center counts from 10 down to 2. It selects the candidate with the most acceptable clusters; ties retain the earlier candidate, which has more centers. Acceptance requires at least 300 members and mean silhouette at least 0.3. After accepted members are removed, d_c is multiplied by 0.9. The process stops when fewer than 300 points remain or five consecutive iterations accept no cluster. The remaining, or residual, points are still unassigned at this stage.

Distances are computed in bounded blocks and stored in a float32 matrix. Cutoff density is evaluated row by row, avoiding an additional floating-point kernel matrix. Silhouettes reuse the distance matrix when the number of labels lies between two and $N - 1$, as required for a valid calculation [24]. Distance storage still grows quadratically with group size, and repeated silhouette calculations can remain costly. These changes reduce extra memory use without removing that scaling problem.

3.7 STEP 2: Wasserstein consensus

Only acceptable local clusters enter consensus. For clusters A and B, the implemented source metric is

$$D(A,B) = \left[\sum_{f=1}^{4} W_1(A_f, B_f)^2\right]^{1/2}. \tag{3.1}$$

Here W_1 is the one-dimensional first Wasserstein distance between the observed distributions of a single descriptor [25]. Equation 3.1 combines these four separate distances; it is not a joint four-dimensional optimal-transport distance. It can miss differences in the relationships between descriptors when each individual distribution remains unchanged.

The reproduction configuration uses raw descriptors, matching the supplied code. The combined distance therefore mixes their scales and units, a choice the paper does not fully explain. Normalized distances belong to a separate extension. Average-linkage clustering merges local clusters and then uses the source's cubic-spline elbow heuristic to choose a cut: smoothing parameter 5, with the maximum second derivative sought in the first 90% of merge steps. The calculation fails explicitly if there are fewer than five local clusters or the merge-height curve is entirely flat, because the heuristic cannot justify a cut in those cases.

3.8 STEP 3: final frame assignment

Each global centroid is the mean normalized feature vector of all accepted local-cluster members assigned to that family. All analysis-window frames, including residuals and frames outside reduced discovery groups, are assigned once to the nearest centroid. Distance ties use deterministic label ordering. Centroids are not iteratively refitted after reassignment: doing so would introduce a k-means refinement absent from the documented baseline.

The final labels are stored with the original frame identities. This lets descriptor statistics, stability comparisons and any later structural mapping refer to exactly the same observations. Their molecular interpretation remains limited by the chosen descriptor representation.

3.9 FES validation and interpolation

The FES parser requires axis names, minimum and maximum coordinates, grid counts, and confirmation that the axes are nonperiodic. It checks the row count, finite values including supplied derivatives, coordinate spacing, unique coordinate pairs, and agreement with the header. Rows need not arrive in grid order. Files with a partial-download extension are rejected before parsing.

Interpolation uses SciPy's linear rectilinear-grid interpolator with bounds checking enabled [26, 27]. Every query must fall inside both axes. A bilinear analytic surface containing a mixed $\alpha\beta$ term provides an exact test, including boundary queries and shuffled rows. Missing cells and out-of-bounds queries are errors; no extrapolation or clipping is permitted.

3.10 Statistical weights

Equation 2.2 is evaluated in log space. Within replica r, normalized weights are

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \operatorname{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \frac{\widetilde{w}_{ri}}{3}. \tag{3.2}$$

Equation 3.2 gives each replica the same total weight, regardless of the arbitrary log-weight offset or differences in raw weight sums. The raw exponential is saved only as a diagnostic. If it is too large or too small to represent numerically, the output flags overflow or underflow; normalized calculations still use log weights.

The intended frame table contains replica, source row, file line, frame index, time, physical CVs, interpolated free energy, raw log weight, raw weight/status, within-replica weight, pooled weight, and

final label. No such real weighted table is reported in this release because FES energy-unit and construction provenance remain unverified.

3.11 Weighted cluster characterization

Summing the pooled weights of a cluster's members gives its relevance. The same member weights give the mean and width of each descriptor within that cluster. The width is the square root of the weighted central second moment and describes how much the members vary. The replica standard error of the mean (SEM) is calculated from the separate replica populations and describes uncertainty in their average relevance.

The uncertainty extension keeps the fitted labels and supplied FES fixed while resampling blocks of consecutive observations independently within each replica. It renormalizes the selected weights and recalculates populations for each draw. Any resulting intervals would apply only to that fixed partition and FES. They would not include uncertainty from reconstructing the FES, selecting a model or changing the clustering seed. The output records this limitation.

3.12 Software environment and reproducibility

The modular implementation uses NumPy, SciPy, pandas, Matplotlib and scikit-learn for numerical analysis, data handling, plotting and machine learning [27–29]. A locked environment records the installed versions separately from the published simulation versions. Snakemake specifies which stages depend on earlier outputs; command-line interface (CLI) commands also run these stages without notebooks [4, 30].

Each run begins with a manifest recording its chosen parameters. A supervisor process records the analysis worker's status and measures memory use across its process tree every 0.1 s, enforcing the configured runtime and memory limits. Successful, failed, and blocked runs have distinct final statuses. Figures can be rebuilt only after a successful run and a check of its output hashes. This record links each figure to the calculation and data that produced it.

4 Reproduction and scientific code audit

4.1 Original implementation

The supplied implementation puts input reading, replica separation, plotting, clustering and summary calculations in one script. Parsing the source identifies an Italian sentence without a comment marker at line 437, causing a syntax error before the analysis can run. The original is preserved exactly, including this defect, so that the audit can be checked against the supplied file.

The audit traces each calculation from its input to the quantity it would report. Reading the source shows what the script attempts to do, but no original execution manifest or complete numerical output set was located. Where the code differs from the paper, the method map records both versions and explains the correction made in the modular implementation.

4.2 Identified methodological issues

The original parser silently takes the first three segments separated by time resets and skips certain malformed lines. The new parser checks both the segment count and each row. Original group generation depends on a fixed list of 17 seeds, which would run out with smaller groups. The replacement uses local RNG streams and preserves every frame's identity. It makes the sampling reproducible but changes the exact subsets, so it is not claimed to reproduce the legacy script bit for bit.

The handling of residual frames changes which observations define a family. In the supplied code, rejected frames receive a generic label and every label enters consensus. Frames that failed the acceptance criteria can therefore influence the global centers. The modular version uses only accepted local clusters to discover families, then assigns the residual frames in STEP 3. Residuals still contribute to the final population without being treated as an accepted local cluster.

The original FES reader ignores the header information needed to check completeness and permits missing grid values with a warning. The weight function uses the nearest grid point and selects a boundary value for out-of-range queries. These choices can produce plausible-looking weights from invalid inputs. The replacement rejects incomplete grids and uses bilinear interpolation only within the grid bounds. The original partial download ended inside a numerical row and was rejected. The complete replacement received on 7 October passed grid validation; the earlier partial file remains unusable.

The original analysis weights cluster relevance but uses ordinary arithmetic means and standard deviations for the structural descriptors. Those summaries describe a different distribution from the weighted one specified in the paper's figure captions. The revised routines use the same weights for relevance and descriptor summaries within each cluster, and normalize replicas separately before pooling. Distribution width, replica SEM and bootstrap intervals remain separate quantities.

Scientific audit note

Correcting a calculation does not validate its input data or establish that the original simulation converged. The complete grid now passes validation, but its energy unit and construction procedure still need clarification before reweighting the real data.

4.3 Scientific V2 corrections: evidence unavailable

The requested intermediate file `CLUSTERING40_scientific_v2.py` and its changelog were not located. The method map therefore marks that implementation as unavailable. None of the corrections in this report is attributed to the missing version. If supplied later, it should be checked by its hash, preserved unchanged, and compared with both the original and the modular implementation.

4.4 Verification and numerical tests

The current suite passed 36 tests, including strict parsing, interval boundaries, normalization, analytic bilinear interpolation, incomplete and duplicate grids, finite/bounds checks, weight sign, free-energy-offset invariance, equal replica masses, weighted moments, residual handling, final assignment, label-permutation agreement, and bootstrap separation of replicas. A seeded autoregressive fixture checks that the correlation estimator behaves plausibly for a known stationary process.

A complete synthetic test dataset exercises cluster discovery, consensus, all-frame assignment, final-bias weights, population tables, conditional bootstrap intervals and PDF generation. Its generated descriptors and analytic surface are used only for testing, not as Aβ40 observations. A further test checks repeatability by fitting the GMM extension to synthetic, separated observations. Tests with renamed labels and missing families check that the comparisons behave as intended. An archive-provenance test also checks that a source release cannot inherit the Git commit of an unrelated enclosing repository. These tests check numerical behavior; they cannot establish that a cluster represents a biological family.

4.5 Remaining methodological ambiguities

The main unresolved choices are the exact descriptor scaling in STEP 2, how the replica-averaged FES was constructed, and how the published cluster widths and uncertainties were defined. The supplied code also gives more detail than the paper about density ties and spline-based elbow selection. The repository identifies these choices as based on the supplied source; it does not imply that the paper fully specifies them.

These choices affect how the analysis is run and how its results can be compared. Raw Wasserstein scaling remains the reproduction setting; normalized scaling has a separate extension configuration. The complete numerical grid does not resolve how the FES was constructed. Any future difference in populations will need to be checked against these analysis choices before it is given a biological interpretation.

5 RobustCluster-Aβ: robustness extension

5.1 Temporal correlation and effective sampling

For each replica and baseline descriptor, the extension subtracts the mean and computes autocorrelation using an FFT-based convolution. Covariances use the common $1/N$ normalization. It pairs adjacent terms at positive lags and stops at the first pair whose sum is nonpositive. Each retained pair sum is then limited to the smallest pair sum seen so far. This gives the diagnostic

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5.1}$$

In eq. (5.1), $\mathscr{K}$ denotes the lag contributions retained by this procedure; the implementation saves the lag at which summation stops. The convention includes the zero-lag half term in τ_{int}. Some definitions omit this term, so a reported "correlation time" needs an explicit convention.

Three checks help assess the estimate: whether a stopping lag was found, whether $N \geq 50g$, and whether the absolute difference between first- and second-half means is at most 0.25 of the full-series standard deviation. The result is flagged if any check fails. These practical checks identify estimates with weak support from the available series; they are not a formal test that the process is stationary. More generally, an uncertainty estimate must be supported by the data and its sampling assumptions [11, 12].

Weight concentration is assessed separately by

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{5.2}$$

Equation 5.2 describes how unequal weights reduce the effective number of contributing observations, without correcting for dependence between them. It is not multiplied by, or automatically divided by, the correlation ESS. Combining the two effects would require a dependence calculation for the particular weighted observable and estimator. No Kish ESS is reported for the real data because their statistical weights have not been validated.

5.2 Cluster stability across seeds

Random grouping affects which local clusters are found, so the planned DPC seed experiment repeats discovery, consensus cutting and final assignment together. Simply renaming the labels of a fixed partition would not test this variation. The smoke profile uses two seeds, the separate pilot uses five, and the full profile, which has not been run, defaults to 20, with support for a 20–50-seed study.

Partitions are compared using the same source frames. ARI measures agreement regardless of the names assigned to clusters [31]. Families are then matched across runs to maximize their overlap, allowing Jaccard overlap, population variation and centroid movement to be summarized. Families without a match remain in the comparison, including those that appear or disappear. Considering only matched families could hide changes in cluster number even when those families look stable.

To compare centroids from different representations, the analysis uses the original four descriptors rescaled by the common baseline ranges. It therefore does not compare a PCA-space centroid directly with a raw-space centroid. Population variation across seeds measures sensitivity to the algorithm's random choices on the same ensemble. It differs from sampling uncertainty between independent replicas.

5.3 Feature redundancy and PCA

The completed lightweight diagnostics calculate the pooled Pearson correlation matrix and PCA after standardizing each baseline feature to zero mean and unit variance. The explained-variance spectrum shows how much of the variation in the observed descriptors lies along each principal component. It does not measure thermodynamic relevance or identify a slow physical mode.

The alternative clustering configurations retain all principal components, with optional whitening to scale each component to unit variance. Keeping all components makes PCA an orthogonal rotation of the standardized variables, preserving their Euclidean distances. Any difference from the min–max baseline would therefore come from standardization, whitening, numerical effects or a later analysis choice, not from discarding dimensions. This distinction matters when interpreting a future change in clustering.

5.4 Independent GMM feature-ablation experiment

The completed original experiment compares the four baseline descriptors with n_α, n_β and R_g after removing SASA. It uses Gaussian mixtures in scalar descriptor space, a different model from both the published DPC pipeline and coordinate-based ShapeGMM. For a standardized observation $\mathbf{x}$,

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{5.3}$$

Equation 5.3 approximates the sampled descriptor distribution. Its mixture coefficients are not equilibrium statistical weights. The implementation uses scikit-learn [29]; each component has a full covariance matrix, allowing the model to represent correlations between its features.

The experiment uses one fixed, replica-balanced fitting sample: 1,000 frames from each replica, selected with sampling seed 20261007. Feature means and standard deviations are fitted on those 3,000 rows and reused for the shared features in both representations. At each $K = 2, \ldots, 6$, five seeds (42–46) initialize separate models, each with a 500-iteration limit, tolerance 10^{-3} and covariance regularization 10^{-6}. Using one initialization per fit exposes variation between local solutions instead of selecting the best of several starts. The protocol was committed before execution.

Every fitted model assigns a label to all 300,003 observations, including those used for fitting. ARI compares the four- and three-feature assignments at each seed and component count; the ten seed pairs are also compared within each representation. Matching clusters by maximum overlap shows which labels correspond across runs. The analysis records Jaccard overlap, unmatched families, and centroid shifts measured in the shared standardized three-feature space. It also records raw label fractions for each replica. These comparisons describe the same data used in the analysis; they do not test prediction on held-out data, and the ranges across runs are not confidence intervals.

Keeping the fitting subset fixed isolates the effects of feature representation and initialization. The experiment does not measure uncertainty from selecting that subset or remove temporal dependence. BIC is saved only for comparisons within a feature space; it is not compared across dimensions or used to identify a physical number of states. The planned DPC ablations remain separate experiments that require their own input and execution records.

5.5 Wasserstein scaling sensitivity

The raw consensus distance and the normalized alternative both combine separate distances for each descriptor; only the scaling changes. This tests the effect of one unresolved implementation choice. Raw SASA and radius-of-gyration values have different units and numerical ranges, so the variable with the larger range can have a disproportionate influence on the combined distance.

The normalized comparison would measure how scaling affects the result, but would not establish which implementation the original authors intended. Joint distances, sliced Wasserstein, energy distance and MMD could test other aspects of similarity between distributions in later studies. None replaces the current baseline.

5.6 Reweighting sensitivity

The optional comparison keeps the fitted partition fixed and compares the final-bias estimator with weights derived from verified `metad.rbias`. It examines the reweighted distributions of all four baseline

descriptors and family relevance. With the labels unchanged, any differences can be traced to weighting rather than to changes in the partition.

The descriptor file contains a column named `metad.rbias`, but the original PLUMED configuration and evidence for its energy unit are missing. Input validation therefore blocks this optional comparison. The implementation keeps the estimators separate, as described in the primary literature and official PLUMED reference [16–18].

5.7 Alternative density-based clustering

DPA would provide a useful benchmark because it estimates density and identifies significant peaks differently [21]. The comparison should examine family number, location, population and correspondence using the same frames and validated weights. No DPA fit has been performed in this project.

Weighted ShapeGMM can analyze biased structural ensembles [22]. Applying it here would require atomic structures, consistent atom selections and alignment, and a verified link between structures and descriptor rows. The structures cannot be recovered from scalar descriptors alone, so this comparison remains deferred.

5.8 Optional external validation

An Aβ40 ensemble constrained by experimental data could provide an independent comparison of global compaction and secondary-structure behavior [7]. Before comparing values, the sequence, temperature, weights, atom selections and definitions of each observable must be checked. A mass-weighted R_g and a Cα-only R_g, for example, need not have directly comparable distributions.

No external ensemble has been processed in this project. The assessment identifies the relevant study and the conditions needed for a valid comparison. Testing agreement with NMR/SAXS-constrained descriptors remains future work.

6 Results

6.1 Available evidence and execution status

The input checks found three segments of 100 001 rows, each at the 20 ps spacing stated in the source. The table had the expected columns, finite values and interval coverage. It can therefore be used for the descriptor calculations below. These checks alone cannot show that the original simulation converged or that each descriptor was calculated correctly.

Run `20260921T011959-583b4421` produced the descriptor results in 2.28 s, with sampled peak worker-process memory of 418.2 MiB. These measurements apply to this small local calculation; they do not predict the cost of full DPC. Table 1 lists the completed diagnostics, FES check and GMM experiment alongside the analyses that remain blocked.

Analysis	Evidence status
Input validation	Completed on the supplied Aβ40 descriptor table
Correlation matrix / PCA	Completed; unweighted biased-sample diagnostics
Autocorrelations / provisional ESS	Completed; every estimate flagged as unreliable
FES grid and interpolation	Completed; units/construction remain unverified
Weighted paper reproduction	Blocked: unit/construction provenance
GMM seed / SASA-ablation pilot	Completed; biased-sample geometric comparison
DPC seed / feature / scaling studies	Not completed; separate reproduction prerequisites
Time-dependent weight comparison	Blocked: FES and rbias unit/provenance requirements
Molecular representatives / ShapeGMM	Blocked: coordinate trajectory and mapping absent
External validation / Leonardo jobs	Not performed

Table 1: Scientific evidence available at release. Tests on synthetic fixtures are excluded from the empirical analysis statuses.

6.2 Descriptor distributions and redundancy

Across the pooled data, the Pearson correlation is 0.848 for Rg and SASA and -0.456 for the two secondary-structure descriptors. The strong Rg–SASA correlation motivated the feature-removal experiment. Both descriptors describe aspects of molecular dimensions, but their correlation does not tell us whether removing one will change the label assigned to a frame.

The first two standardized principal components explain 51.8% and 36.3% of total variance, respectively, or 88.1% together (fig. 3). Most of the variation is therefore captured by two directions in feature space. The clustering comparison checks whether distinctions with less variance still affect the fitted groups.

Figure 4 compares the replicas with the same bin edges for each descriptor. Each histogram is divided by its own frame count and bin width. The curves show differences between the recorded samples, but no equilibrium correction has been applied, so they cannot establish differences in equilibrium populations.

6.3 Temporal correlation and provisional ESS

The provisional ESS values range from 3.2 to 28.7 across the 12 replica–descriptor series, compared with 100 001 saved frames in each series. All 12 estimates fail at least one of the reliability checks defined earlier. The results point to strong dependence and possible changes over time, but they should not be read as precise numbers of independent equilibrium conformations.

Table 2 reports these estimates and the difference in means between the two halves of each series. Several descriptors show a substantial shift. Together with the failed reliability checks, this gives little support for treating frames as independent when estimating uncertainty. A comparison with the published

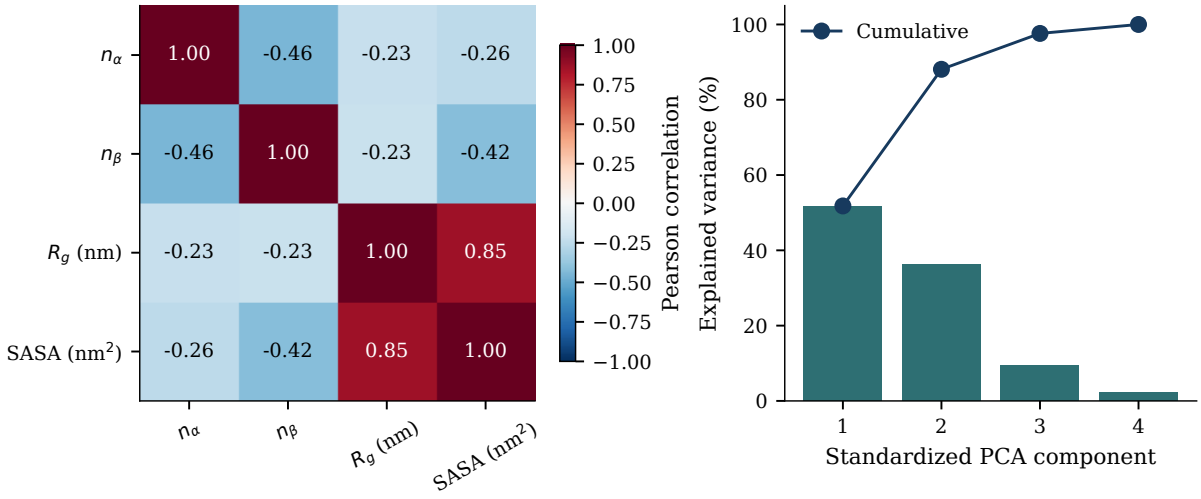


Figure 3: Computed diagnostics for the pooled, unweighted Aβ40 descriptor sample. Left: Pearson correlations of the four baseline descriptors. Right: explained variance of PCA after per-feature standardization, with cumulative variance shown by the line. These are geometric summaries of the biased sample and do not represent reweighted equilibrium probabilities.

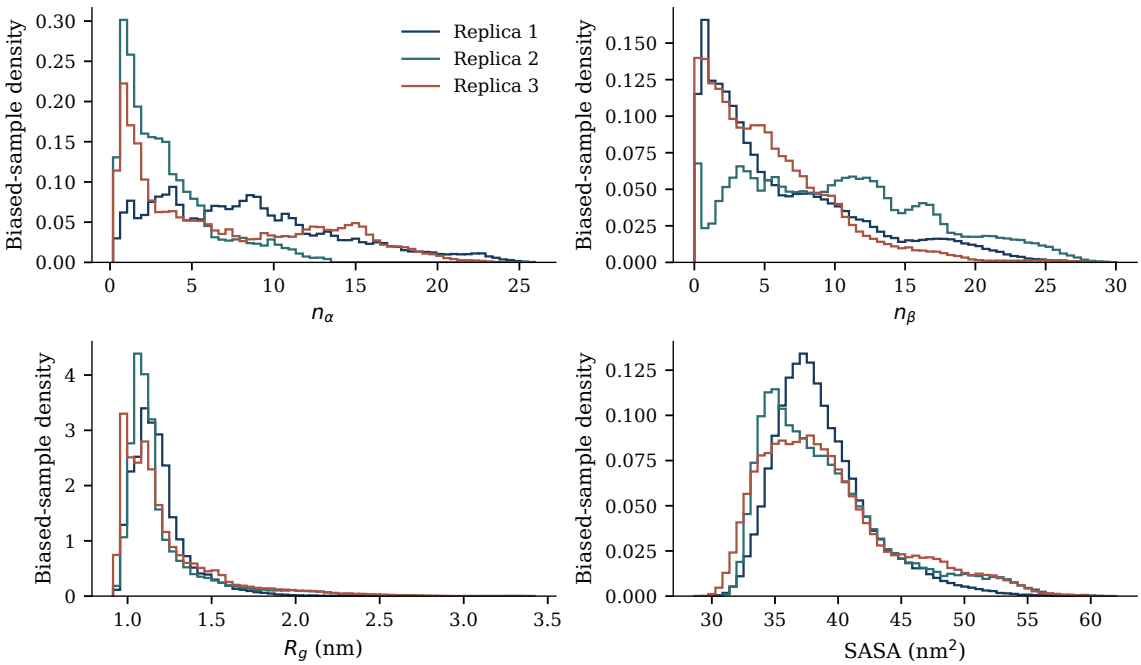


Figure 4: Actual per-replica descriptor histograms over the supplied interval. Curves use common bin edges for each descriptor and independent density normalization. No final-bias or time-dependent statistical weights have been applied. Units follow the original script's labels and remain a recorded upstream-provenance assumption.

convergence assessment would need the full bias history and the original diagnostics, which are not available here.

Replica	Descriptor	g	τ_{int} (ns)	$N_{\mathrm{eff,corr}}$	Half shift / SD
1	n_α	22067	220.7	4.5	1.28
1	n_β	11079	110.8	9.0	0.83
1	R_g	3679	36.8	27.2	0.27
1	SASA	3490	34.9	28.7	0.36
2	n_α	6259	62.6	16.0	0.37
2	n_β	10843	108.4	9.2	0.41
2	R_g	3951	39.5	25.3	0.33
2	SASA	5325	53.2	18.8	0.17
3	n_α	31742	317.4	3.2	1.61
3	n_β	5880	58.8	17.0	0.71
3	R_g	6259	62.6	16.0	0.88
3	SASA	9958	99.6	10.0	0.87

Table 2: Computed correlation diagnostics. All entries are flagged as unreliable under the stated heuristics. The time scale is measured along the biased trajectory, using the source-labelled sampling interval. g is statistical inefficiency; the final column is the absolute difference of half-window means divided by the full-series standard deviation. No physical relaxation-time interpretation is made.

6.4 Complete FES validation

The recovered file contains exactly 2,801 by 1,301 grid points: 3,644,101 rows with five finite columns. There are no duplicate coordinates, and every expected pair of grid coordinates is present. The nonperiodic axes span 0–28 and 0–13 at spacing 0.01. All 300,003 converted descriptor queries are inside the grid, and bilinear interpolation returns a finite value for each one. This confirms that the grid can be read and queried. It does not resolve the missing energy unit; the source script's use of kJ/mol is not enough to verify that unit for the file.

6.5 Representation and initialization sensitivity

All 50 GMM fits reached the chosen numerical convergence criterion. The run took 7.01 s, with a sampled process-tree peak of 660.2 MiB. Even so, the frame assignments differ between component counts and initializations (fig. 6). An algorithm can finish fitting without giving the same partition in every run. At three components, ARI between the four-feature and SASA-removed fits lies between 0.862 and

Components	Median ablation ARI	Observed range
2	0.690	0.188–0.868
3	0.864	0.862–0.868
4	0.607	0.548–0.622
5	0.531	0.382–0.696
6	0.457	0.294–0.461

Table 3: Observed representation agreement across five paired initializations. Component count was varied by design; the table does not identify an optimal biological family count.

0.868. After matching labels by maximum overlap, approximately 95.3% of frames keep corresponding labels (median across seeds). Agreement between seeds is also high: 0.994–1.000 with four descriptors and 0.988–1.000 without SASA. At two components, within-space ARI falls as low as 0.011 and 0.067, respectively. One initialization would not reveal this variation. Figure 7 shows the full comparison matrices.

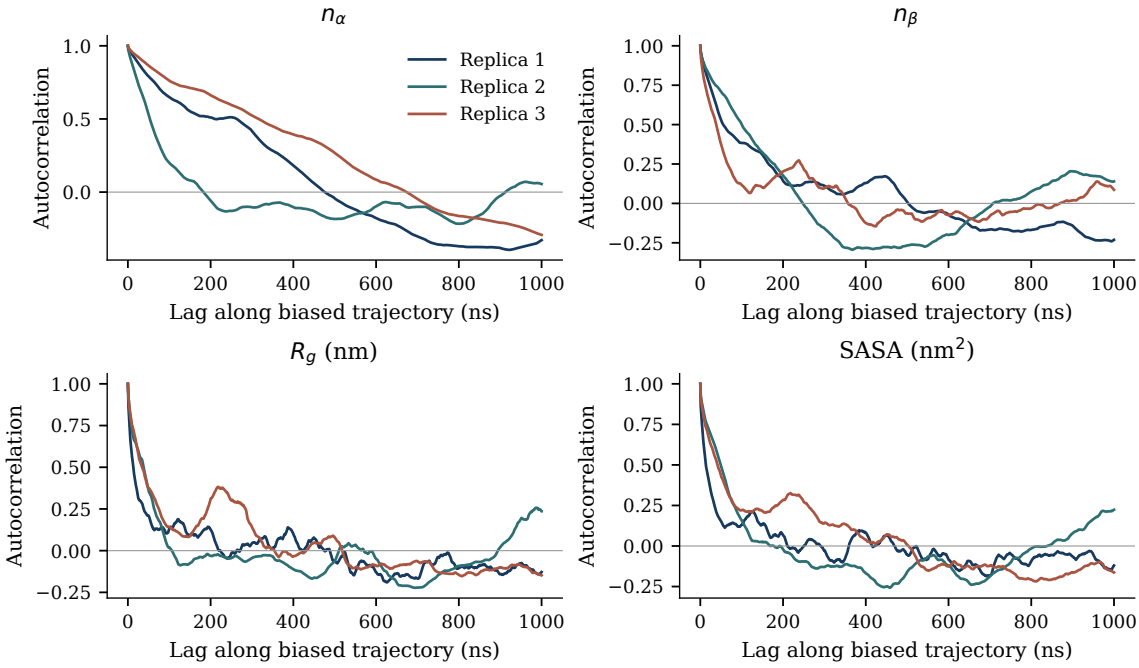


Figure 5: Autocorrelation functions computed separately for each replica and descriptor. The plotted curves cover the first half of the available lag window with a reduced plotting density; the estimator operates on the full, un-decimated ACF sequence. Long persistence and changing behavior across replicas reinforce the need for cautious uncertainty analysis. Lag is biased trajectory time, not recovered physical time.

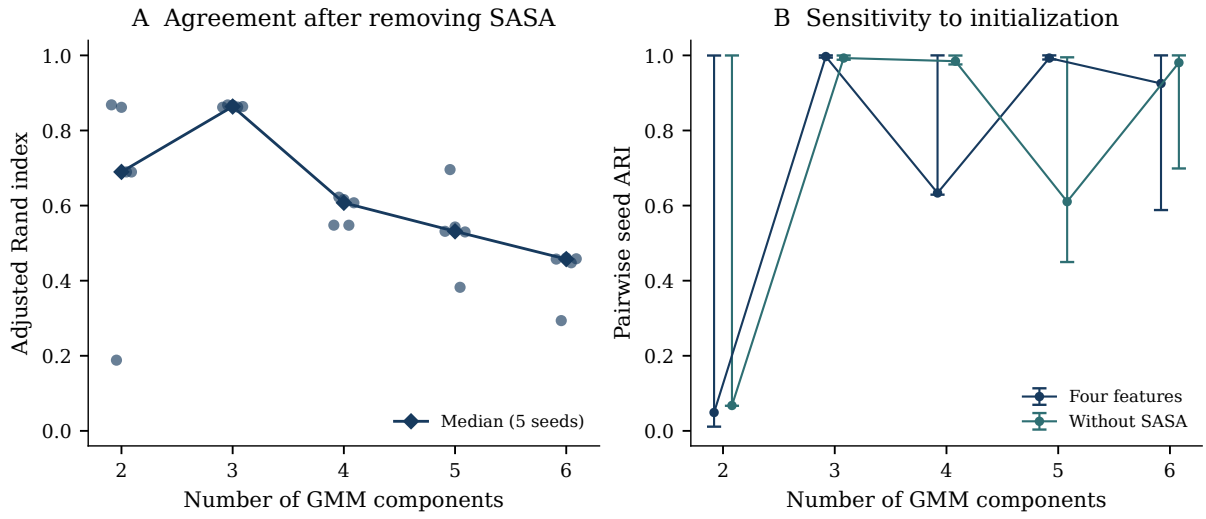


Figure 6: Independent GMM sensitivity experiment on the biased descriptor sample. Left: four-feature versus SASA-ablated assignments at each of five seeds; diamonds mark medians. Right: within-space ARI over ten seed pairs; bars span observed minima and maxima, not confidence intervals. The fit subset is fixed across all models and evaluation includes fit rows.

The three-component fits show the clearest agreement after removing SASA. At the other tested counts, removing SASA has a larger effect or gives results that vary more between seeds. The conclusion is therefore tied to the tested model size and fitting procedure. It does not justify dropping SASA from the published baseline. These are comparisons of labels in the biased sample, with no equilibrium weighting.

6.6 Analyses remaining unavailable

Equilibrium cluster relevance, weighted descriptions of the families and the planned paper-DPC robustness studies are still unavailable. Molecular snapshots also require atomic coordinates, which the descriptor table does not contain. The completed GMM experiment answers a separate question about the descriptors and does not replace those analyses.

7 Discussion

7.1 What the completed experiments support

The feature-removal experiment shows why the correlation matrix was not enough to decide whether SASA could be omitted. Rg and SASA are strongly correlated in the pooled sample, yet the fitted groups can change when SASA is removed. Three-component mixtures agree well both between feature sets and between initializations. Other component counts are less consistent. The effect of removing a feature therefore depends on the model, not just on its correlation with another feature.

This comparison has a clear limit: every model was fitted to the same selected observations. It tests changes in features and initialization, but not the effect of choosing another fitting sample. Random frame selection also does not remove the dependence already present in the trajectory. Every replica–descriptor series fails at least one correlation-estimator reliability check, so the provisional ESS values cannot be treated as reliable ensemble sizes. Repeating the fits with new subsets and with time blocks would help test how much the sample choice matters [12].

7.2 Relation to the published study

The code implements the published analysis design, but the reweighted conformational families and their populations have not yet been reproduced [2]. The GMM experiment compares analysis choices in descriptor space. It does not test the paper's biological conclusions: neither a correlation nor stable Gaussian labels can establish aggregation rates, toxicity or metastable states. Comparing these results with an independently constrained ensemble would also require compatible structures and definitions of the measured properties [7].

7.3 Limitations and methodological implications

There are two further questions about the models. A converged fit may be a local solution, and agreement across seeds tells us only about the starting points tested. Also, repeatable groups of scalar descriptors may not correspond to distinct physical states. If trajectories become available, representative structures, contact maps and other descriptors could help examine that connection. In the separate consensus baseline, comparing one descriptor at a time can also miss differences in how descriptors vary together.

The FES introduces a separate uncertainty. The grid is complete and covers all query points, but its energy units, the way replicas were combined and the convergence of the bias remain unknown. Once weights can be calculated, a few large weights could make estimates depend strongly on a small part of the sample. That effect needs to be assessed alongside time correlation and initialization. With only three replicas, there is also limited information for estimating uncertainty from differences between replicas.

Several parts of the work can already be checked by another reader. The saved labels are enough to recalculate GMM agreement without refitting the models, and the saved tables can regenerate the descriptor figures. Equilibrium family populations require the missing FES information and another calculation. Keeping these tasks separate makes the current results reusable while showing exactly what is still needed [6, 32].

8 Internship outcomes and skills

8.1 Learning sequence and practical objectives

The internship connected Data Analysis methods with molecular simulations. The learning programme began with amino-acid properties, peptide bonds, protein structure and intrinsic disorder. It then covered force fields, molecular dynamics and dihedral angles through supplied slides and readings. The final assignment brought these topics together: reproduce a clustering analysis, examine parameter changes and carry out an independent analysis using methods from the degree.

Two ideas were especially relevant to the analysis. Consecutive rows are saved configurations, but they are not necessarily independent observations. Similarly, counting the frames in a group gives its frequency in the saved sample; finding its equilibrium probability requires valid statistical weights. These familiar statistical questions need careful interpretation when the data come from a biased simulation.

Learning to use GROMACS was one of the more challenging parts of the internship for me. I found the process demanding at first, but I also enjoyed becoming more familiar with the software.

8.2 Demonstrable analysis and software outputs

The completed work includes the code and input checks, modular analysis routines, 36 passing tests, descriptor diagnostics, validation of the full FES grid and the GMM sensitivity experiment. The last of these turns the observed Rg–SASA correlation into a question that can be tested by comparing labels. Keeping the results for every component count and initialization shows where assignments are stable and where they change.

Feature scaling is one example of why an implementation detail matters. Once the mean and scale have been fitted on the chosen sample, the model must reuse them for later observations. The following searchable excerpt comes directly from the analysis code.

01 Fit the geometry once RobustCluster-Aβ / Python

```python
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

The function rejects a feature that is constant in the fitting sample and returns the parameters used for scaling. The test supplies a large evaluation value and checks that it does not change the fitted mean or scale. In this way, adding observations for prediction cannot silently change the coordinate system used by the model.

8.3 Companion projects: VMD MCP and GROMACS MCP

Two additional software projects are publicly available: VMD MCP and GROMACS MCP. Both use the Model Context Protocol (MCP) to make molecular-software operations available to a client through structured inputs and outputs. Their code shows how a request is turned into a calculation. The underlying software and the supplied data still determine whether that calculation is scientifically meaningful.

VMD MCP constructs Tcl commands for operations such as per-frame radius of gyration, RMSD, SASA and rendering. The gyration function passes the selected atoms and mass-weighting option to VMD, then parses frame-indexed values. GROMACS MCP includes helpers for simulation preparation and analysis and a background-run interface that constructs a command, validates paths and records output. Genuine source screenshots and immutable commit links appear in figs. 8 and 9.

These projects work with molecular software, while RobustCluster starts from an existing descriptor table. There is no evidence that either companion project generated the supplied Aβ40 trajectories. Their outputs also need care before comparison: the inspected VMD gyration function documents angstroms, whereas the descriptor table is labelled in nanometres. A comparison would need checked units and compatible atom selections. The repository histories retain contributor attribution.

8.4 Reproducibility and computational scope

The software work is documented through preserved inputs, separate configuration files, analytic tests, failure records and figures generated from saved tables. A manifest records which data and settings produced a result. It also distinguishes a calculation that failed or could not start from a successful calculation that returned zero. These records help explain the analysis without relying on the code alone.

All calculations newly reported here were local analyses of descriptors. The project includes a configurable Leonardo submission template and references to the official documentation [33], but there are no Leonardo job records. The reported computational work therefore does not include HPC execution, setting up the molecular simulations or measuring physical kinetic rates.

9 Conclusions and future work

This project checked the supplied Aβ40 analysis code and data, built tested analysis routines and compared Gaussian-mixture fits with and without SASA. The audit found specific problems in execution, residual-cluster handling, FES interpolation, replica pooling and weighted summaries. The modular implementation corrects these problems while preserving the original files and testing the revised operations.

The GMM experiment shows that removing SASA has different effects at different component counts and initializations, despite the strong Rg–SASA correlation. The three-component groups are largely preserved in the tested fits. That finding does not extend automatically to other model sizes or fitting samples. The time-correlation checks also show why the number of saved frames should not be used as the number of independent observations; their flagged ESS values remain rough warning indicators rather than precise counts.

To continue the published-workflow reproduction, the FES energy units and the procedure used to combine replicas must first be verified. A small smoke run can then check the pipeline on real data and measure the resources needed for larger runs with the published parameters. Repeating those runs across seeds, features and Wasserstein scaling choices would show how much the families change when the same frames are analysed differently.

Verified bias-offset data would allow a comparison with time-dependent reweighting. Atomic trajectories and a reliable frame map would allow representative structures and coordinate-based comparisons with tools such as MDAnalysis and ShapeGMM [22, 34]. Any external experimental-ensemble comparison would first need compatible definitions of the measured properties and permission to use the data.

The work therefore provides a result that can be checked from the saved descriptor assignments, together with code for the remaining analysis. Its main finding is that feature correlation alone does not decide whether a descriptor can be removed from clustering. Establishing equilibrium conformational families will require the missing FES information and the calculations that depend on it.

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A Full parameter table

Table 4: Reproduction settings and execution profiles. These parameters are implemented but do not imply a completed empirical run.

Parameter	Value	Evidence / role
Replicas	3	Primary paper; observed table segmentation
Analysis interval	1–3 μs	Published baseline; inclusive source-labelled ps boundaries
Temperature	310 K	Published weighting parameter; no new MD run
Bias factor γ	10	Published weighting parameter
Molar gas constant	0.008314462618 kJ mol^{-1} K^{-1}	Implementation unit convention
CV conversions	$\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$	Published descriptor scaling
Baseline features	n_α, n_β, R_g, SASA	Pooled min–max normalization
Initial d_c	1.3	Published baseline
Cutoff multiplier	0.9	Published 10% reduction
Center-count scan	10 down to 2	Published baseline
Minimum cluster size	300 frames	Published acceptance criterion
Minimum mean silhouette	0.3	Published acceptance criterion
Failure stopping count	5	Consecutive iterations with no accepted cluster
Distance storage	float32	Source-informed; blockwise construction
Consensus	Average linkage	Marginal Wasserstein norm
Elbow smoothing	$s = 5$	Supplied source heuristic
Elbow search range	First 90% of merges	Supplied source heuristic
Final assignment	One nearest-centroid pass	No iterative refitting
Master seed	42	Default project configuration; recorded per run
Smoke / medium groups	3,000 / 10,000	Reduced execution profiles
Smoke / medium discovery groups	2 / 3	Not full-production evidence
Full group size	About 30,000	All groups; full execution disabled
Seed counts	2 smoke; 5 pilot/medium; 20 full default	Full robustness configurable; not performed
Bootstrap draws	500	Conditional moving-block extension; not performed on real weights
Resource policy	Bounded sequential local workers	Full execution held for review

Completed descriptor-only GMM pilot

The following settings belong to the successful independent experiment, not to the published DPC baseline. The fit sample and initializations were recorded before execution.

Table 5: Settings of the completed GMM experiment, as recorded in its run manifest.

Parameter	Value	Evidence / role
Fit sample	1,000 per replica	3,000 rows; shared across models
Sampling seed	20261007	Local NumPy generator
Feature spaces	Four descriptors; omit SASA	Same scaling for shared features
Scaling	Fit-sample mean and SD	No evaluation-data refit
Components	2, 3, 4, 5, 6	All counts retained in comparison
Initialization seeds	42, 43, 44, 45, 46	Paired across representations
Initializations per fit	1	50 total fitted models
Covariance	Full	Regularization 10^{-6}
Convergence settings	500 iterations; tolerance 10^{-3}	All fits converged numerically
Evaluation	300,003 rows	Includes fit rows; descriptive comparison
Resource limits	600 s; 4 GiB	Sequential local worker

B Mathematical derivations and estimator checks

Why the sign of the final-bias correction is negative in FES form

Under a static bias V, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in eq. (2.2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica r. The pooled expectation of A is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses A equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0, 1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and uv. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

C Additional diagnostics and interpretation

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a plotting subsample, not the series used to estimate g. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

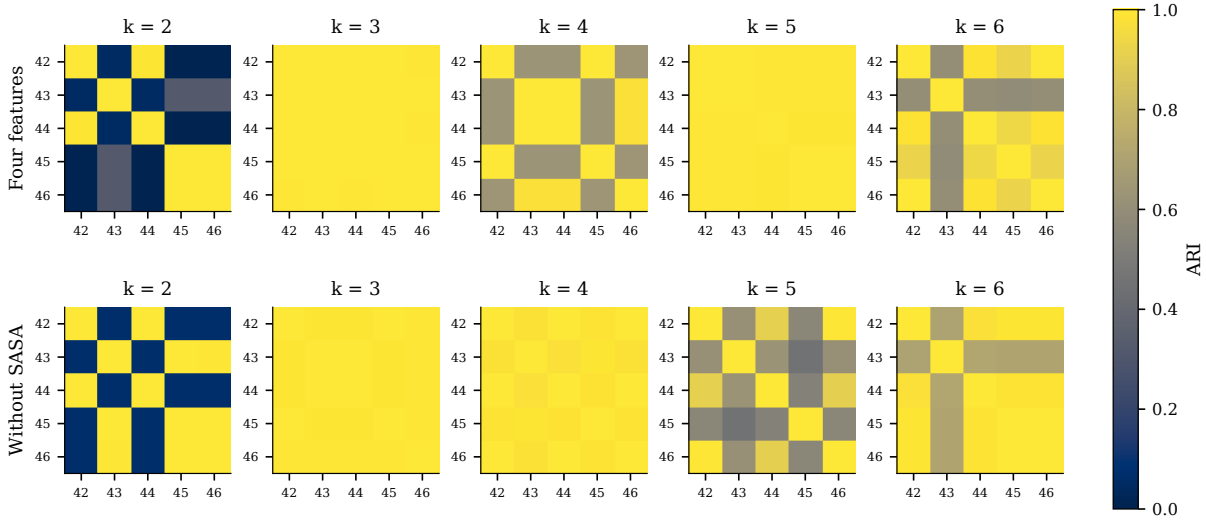


Figure 7: All pairwise seed ARIs from the GMM pilot. Columns show each tested component count; rows distinguish the four-feature and SASA-ablated representations. The diagonal is identity. All fits converged, so no entries are omitted. Similar numerical stopping status can coexist with different partitions.

D Code audit summary

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

Codex assisted with software refactoring, numerical checks, document preparation and layout. Reported numerical observations come from recorded Python runs; the illustrated cover and two conceptual editorial artworks were generated with the built-in image-generation tool, and the source-code screenshots are actual browser captures. This assistance is separate from the published methodology and the original supplied materials. The generation tool does not expose a verifiable model-version identifier; no specific Image 2.5 execution claim is made. Exact prompts and image hashes are retained in `docs/ARTWORK_PROMPTS.json` and `report/artwork_provenance.json`. The supplied inspiration image guided the print texture and restrained palette, not the scientific content. The cover typography is part of the generated image; its institutional and personal details were visually checked. The University of Messina logo is a separate image supplied by the student and placed without generative reconstruction.

E Reproducibility instructions

The project source, tests, configurations and this report are available at RobustCluster-Aβ on GitHub. The Python package and LaTeX sources can be inspected separately or downloaded together using GitHub's *Code / Download ZIP* option. The repository guide distinguishes code verification and report regeneration from analyses requiring restricted inputs.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The report imports successful diagnostics, FES-audit, GMM and figure manifests with matching hashes. Its generated editorial artwork and real browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

For later production, establish the recovered FES energy units and construction provenance, run smoke validation, review resource measurements, and create a reviewed full-run configuration. HPC templates require an actual allocation and explicit settings. Do not copy account or partition values from unrelated examples, and preserve job evidence before attributing calculations to Leonardo [33].

F Software and environment table

Component	Version in diagnostic manifest
Python	3.12.6
numpy	2.5.3
scipy	1.18.1
scikit-learn	1.9.1
pandas	2.3.3
matplotlib	3.11.2
robustcluster-abeta	0.1.0

Table 6: Actual software versions for the reported local descriptor diagnostics; the GMM pilot manifest records the same versions. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.

G Companion source-code evidence

The following images are genuine browser captures from public, commit-specific GitHub source pages on 7 October 2026. They document implementation, not simulation execution. The full images are preserved unchanged; short searchable listings in the main text are a separate presentation format.

02 Measure each saved frame — VMD MCP / source lines 250–266

```
250      weight = "weight mass" if mass_weighted else ""
251      tcl = f"""
252  {_load_block(structure, trajectory)}
253  set sel [atomselect $mol "{_tcl_quote(selection)}"]
254  set n [molinfo $mol get numframes]
255  for {{set i 0}} {{$i < $n}} {{incr i}} {{
256    $sel frame $i
257    puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
258  }}
259  quit
260  """
261      res = _run_tcl(tcl)
262      series = [
263          {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
264          for m in res.get("markers", [])
265          if "rgyr" in m
266      ]
```

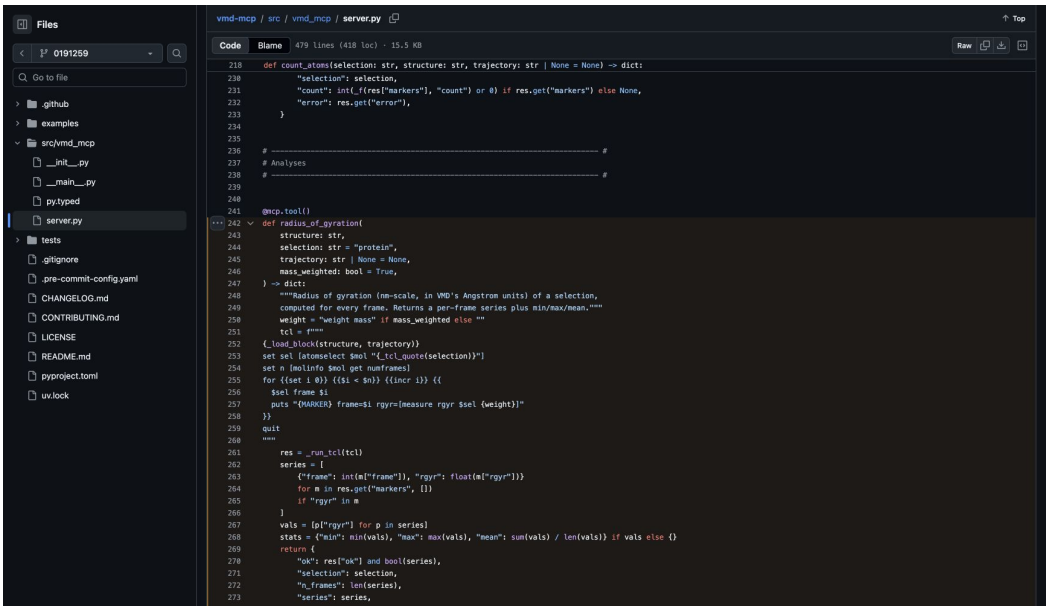


Figure 8: VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. The function's documented unit convention must be checked before comparison with other tools. Open the inspected source.

03 Validate before starting a job GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

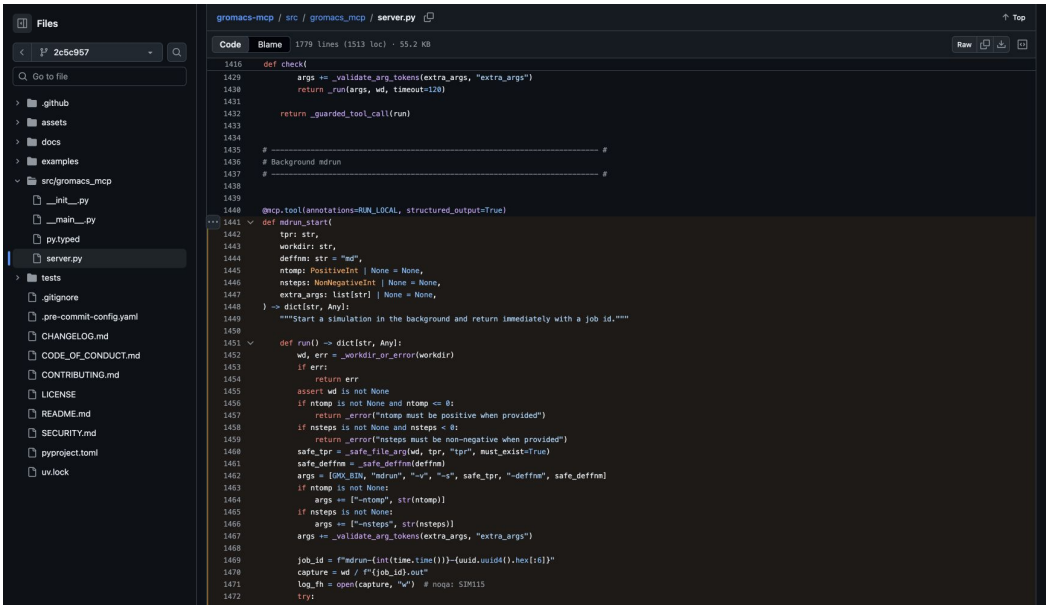


Figure 9: GROMACS MCP source at commit `2c5c957`: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. Open the inspected source.